QP Code 26900067

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU- UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026
FIFTH SEMESTER DISCIPLINE SPECIFIC CORE COURSE (DSC)- MG5DSCBOT301-
PLANT CELL AND MOLECULAR BIOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Part A
Short Answer Questions
Answer Any 10 Questions
Each Question Carries 1 Mark

1. Name the scientists who proposed cell theory.

Ans: Matthias Schleiden and Theodor Schwann proposed the cell theory in 1838 and 1839, respectively. Rudolf Virchow later added that all cells arise from pre-existing cells.

2. Define telomere

Ans: A telomere is the ends of linear chromosomes that have a repetitive DNA sequence associated with proteins and it protects the chromosome from degradation and end-to-end fusion.

3. A cell shrinks and breaks into membrane-bound fragments without affecting neighbouring cells. Identify the process.

Ans: Apoptosis (programmed cell death).

4. State the function of RNA polymerase.

Ans: RNA polymerase synthesizes RNA using a DNA template during transcription.

5. Define RNA splicing.

Ans: RNA splicing is the process of removing introns from a precursor mRNA (pre-mRNA) and joining the exons to produce mature mRNA.

6. Identify which phase of the cell cycle shows the greatest overall cell growth before DNA synthesis begins.

Ans: G_1 phase of interphase.

7. What is the functional role of DNA ligases in DNA replication?

Ans: DNA ligase joins DNA fragments by forming phosphodiester bonds, particularly between Okazaki fragments on the lagging strand.

8. What type of bond holds the two strands of DNA together?

Ans: Hydrogen bonds between complementary nitrogenous bases.

9. Describe the basic structure of the nuclear pore complex.

Ans: The nuclear pore complex is a large protein assembly made of nucleoporins that forms a channel across the double membrane of the nuclear envelope, regulating transport between the nucleus and cytoplasm.

10. What is the process by which an ancestral cell engulfed a bacterium called?

Ans: Endosymbiosis.

11. Who formulated the base-pairing rules for DNA?

Ans: Erwin Chargaff, who established the relationships between the amounts of adenine and thymine and between guanine and cytosine in DNA.

12. What process allows lysosomes to degrade worn-out intracellular organelles?

Ans: Autophagy, in which cellular components are enclosed and delivered to degradative compartments. In plants, the central lytic vacuole performs many equivalent degradative functions.

Part B

Short Essay Questions
Answer any 6 questions
Each Question Carries 1 Mark

13. Explain the significance of conserved genes and proteins, citing suitable examples.

Ans: Conserved genes and proteins are genes and proteins that have remained relatively unchanged during evolution because they perform essential biological functions. They are important tools for understanding cellular functions and evolutionary relationships.

Significance:

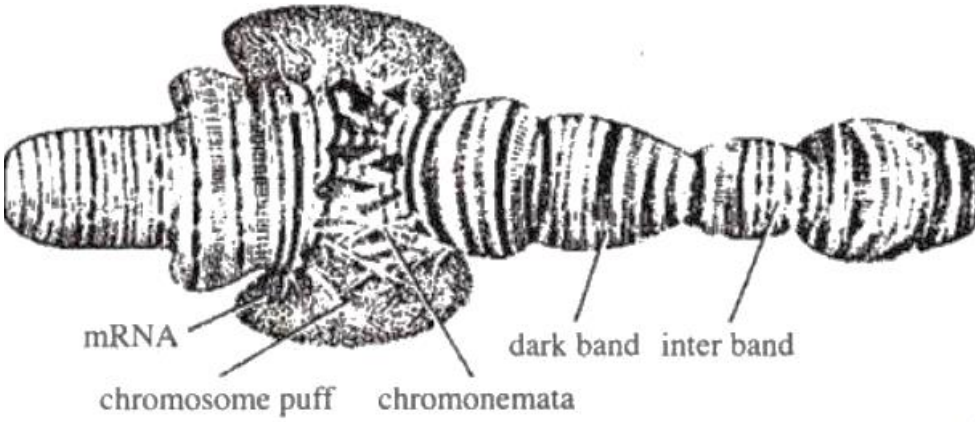
- Essential cellular functions: Conserved genes encode proteins necessary for basic cellular processes.
- Evidence of evolution: Similar genes and proteins in different organisms indicate common ancestry.
- Phylogenetic studies: Conserved sequences help establish evolutionary relationships.
- Identification of organisms: DNA sequence comparisons can help identify species.
- Research applications: Conserved genes provide useful models for studying gene function and disease mechanisms.

Examples:

- Ribosomal RNA genes are conserved because they are essential for protein synthesis.
- Histone proteins are conserved because they package and organize DNA.
- Cytochrome c is involved in cellular respiration and has been used in evolutionary comparisons.
- ATP synthase proteins are conserved because they participate in ATP production.

14. Illustrate and explain the salivary gland chromosome.

Ans: Salivary gland chromosomes, also called polytene chromosomes, are giant chromosomes found in the salivary gland cells of certain dipteran insect larvae, such as Drosophila. They are formed due to the repeated rounds of DNA replication occur in the salivary gland cells without cell division. The replicated chromatids remain aligned, forming giant chromosomes. They show bands and interbands. Dark bands contain densely packed chromatin, while lighter interbands contain less densely packed chromatin. The regions which are at active gene transcription become enlarged due to local chromatin decondensation and are known as chromosomal puffs. These prominent puffs with intense RNA synthesis are known as Balbiani rings. Salivary gland chromosomes were used in studying chromosome organization, gene activity.



15. Describe the wobble hypothesis proposed by Crick.

Ans: The wobble hypothesis, proposed by Francis Crick in 1966, explains how a single tRNA molecule can recognize more than one codon during protein synthesis/ more than one code for the same amino acid. This is known as the degeneracy of the codon.

According to woble hypothesis, the first two bases of the codon generally form specific base pairs with the corresponding bases of the tRNA anticodon but the third base at the codon's 3′ end can show flexibility in pairing with the first base at the anticodon's 5′ end. This is called wobble pairing. Thus explains the degeneracy of the genetic code. Due to Wobble pairing single tRNA can recognize multiple synonymous codons that specify the same amino acid. This reduces the number of different tRNA molecules required for translation.

16. Write a comparative note on heterochromatin and euchromatin.

Ans: Heterochromatin and euchromatin differ in packing, transcriptional activity and functional roles in the regulation of gene expression.

Heterochromatin	Euchromatin
Chromatin Highly condensed	Chromatin less condensed

Darkly stained	Lightly stained
Transcriptionally inactive or less active	transcriptionally active
Structural support, chromosome stability and gene silencing	Expression of genes and regulation of cellular activities
More repetitive DNA	Less repetitive DNA

Heterochromatin can be Constitutive or Facultative heterochromatin.Constitutive heterochromatin remains condensed in most cells commonly found around centromere, telomere regions etc. Facultative heterochromatin can switch between condensed and less-condensed states depending on developmental or cellular conditions.

17. Discuss how the cytoskeleton helps in cell structure and function.

Ans: The cytoskeleton is a dynamic network of protein filaments that provides structural support and essential for intracellular movement, transport and accurate cell division. The major components and functions of cytoskeleton are given below.

- **Microtubules**: Made of tubulin. They maintain cell organization, form the mitotic spindle, and participate in intracellular transport and the structure of cilia and flagella.
- **Microfilaments**: Made mainly of actin. They help maintain cell shape, support cytoplasmic streaming and contribute to cell movement and cytokinesis.
- **Intermediate filaments**: Provide mechanical strength and help resist cellular stress in organisms that possess them. Plant cells do not have the typical cytoplasmic intermediate-filament system characteristic of animal cells.
- **Cell division**: Microtubules organize the spindle that separates chromosomes during mitosis.
- **Intracellular transport**: Cytoskeletal tracks enable motor proteins to transport vesicles and organelles.
- **Plant cell organization**: The cortical cytoskeleton helps guide cellulose synthase complexes and thereby influences the orientation of cellulose microfibrils in the cell wall.

18. What are frame-shift mutations? Explain their significance.

Ans: A frame-shift mutation is a mutation caused by the insertion or deletion of nucleotides in a coding sequence in a number that is not a multiple of three. It changes the reading frame of the genetic code. Frame-shift mutations can have severe effects because they may alter a large portion of a protein.

Insertion: Addition of one or more nucleotides.

Deletion: Removal of one or more nucleotides.

If one nucleotide is inserted or deleted, the grouping of downstream codons changes. This can alter many amino acids and may introduce a premature stop codon.

Significance:

- May produce an abnormal or non-functional protein.
- Can change the amino acid sequence downstream of the mutation.
- May introduce premature termination of translation.
- Can cause inherited disorders or contribute to cancer when important genes are affected.
- Frameshifts can also be useful in genetic research for identifying gene functions.
- Insertions or deletions in multiples of three do not ordinarily shift the reading frame, although they may add or remove amino acids.

19. DNA repair mechanisms in cells are essential for the normal well-being of an organism. Why?

Ans: In every organism DNA is continuously exposed to damage from ultraviolet radiation, chemicals, reactive oxygen species and errors during DNA replication. DNA repair mechanisms correct this damage and preserve genetic stability.

The significance of DNA repair mechanisms include

- Maintenance of genetic information: Repair mechanisms restore damaged DNA and preserve the integrity of the genome.
- Prevention of mutations: They reduce the accumulation of permanent changes in DNA sequences.
- Prevention of cancer: Repair of DNA damage in genes controlling cell growth helps reduce the risk of cancer.
- Normal cell functioning: DNA repair enables accurate replication and expression of genetic information.
- Cell survival: Repair of potentially lethal DNA damage supports cell viability.
- Reproductive and developmental health: DNA repair helps maintain genome integrity in germ cells and developing tissues.
- Ageing and disease prevention: Defects in repair systems can contribute to premature ageing, developmental abnormalities and genetic disorders.

20. Describe the structure and functions of mitochondria in a plant cell.

Ans: Mitochondria are double-membrane-bound organelles that produce most of the ATP required for cellular activities through aerobic respiration. They are often called the powerhouses of the cell. Since they have their own genetic material and ribosome they are considered as semiautonomous organelle.

Structure: The membranes of mitochondria include outer membrane and inner membrane. The outer membrane is smooth membrane enclosing the organelle; contains proteins that permit passage of certain small molecules. The inner membrane is highly selective membrane and it contain proteins in the electron transport chain and ATP synthase. The folds of the inner membrane are called Cristae that increase the surface area available for ATP production.

In mitochondria there is two space system. In between outer membrane and inner membrane their occurs intermembrane space. Inner compartment or the space within the inner membrane

containing enzymes for the link reaction and Krebs cycle, along with mitochondrial DNA and ribosomes. Mitochondria possess their own DNA and ribosomes and can synthesize some of their proteins.

Functions:

- Produces ATP through oxidative phosphorylation.
- Carries out the Krebs cycle and associated respiratory reactions.
- Oxidizes products derived from carbohydrates and other respiratory substrates.
- Participates in metabolic pathways, including aspects of amino acid and lipid metabolism.
- Helps regulate cellular redox balance and responses to oxidative stress.
- Interacts with chloroplasts and other cellular compartments to support plant metabolism.

Part C

Essay Questions

Answer any 1 question · 10 marks

21. Describe the basic chemical composition of the plant cell membrane.

Ans: The plant cell membrane, also called the plasma membrane, is a thin, selectively permeable membrane that separates the cytoplasm from the external environment. It regulates the movement of substances into and out of the cell and helps maintain cellular homeostasis. According to the fluid mosaic model, proposed by Singer and Nicolson in 1972, the membrane consists of a fluid phospholipid bilayer containing proteins and associated carbohydrates.

Basic chemical composition

The major components of plasma membrane are lipids, Proteins and Carbohydrates.

1. Lipids

Lipids form the basic structural framework of the plasma membrane. Phospholipids are the major structural components. Each phospholipid molecule has a hydrophilic phosphate-containing head and hydrophobic fatty acid tails. They arrange themselves into a bilayer, with the heads facing the aqueous environments and the tails directed inward.

Plant membranes contain sterols such as sitosterol, stigmasterol and campesterol. These influence membrane fluidity, stability and permeability.

Glycolipids present in plasmamembrane are Lipids containing carbohydrate groups that contribute to membrane organization and recognition.

Functions of lipids include:

- Form the membrane's basic barrier.

- Maintain membrane fluidity and flexibility.
- Regulate the permeability of the membrane.
- Support the organization and functioning of membrane proteins.

2. Proteins

Proteins are embedded in or associated with the phospholipid bilayer. They include integral membrane proteins and peripheral membrane proteins. Integral proteins are seen embedded within the bilayer; many span the entire membrane. Peripheral proteins are seen associated with the membrane surface or with other membrane proteins.

Functions of proteins include:

- Transport substances through channels, carriers and pumps.
- Act as receptors for signalling molecules.
- Function as enzymes.
- Participate in cell recognition and adhesion.
- Connect the membrane with the cytoskeleton and extracellular structures.

3. Carbohydrates

Carbohydrates occur mainly on the outer surface of the plasma membrane, attached to proteins (Glucoproteins) or lipids (Glycolipids).

Functions of carbohydrates include:

- Help in cell recognition.
- Participate in cell-to-cell interactions.
- Contribute to membrane protection and surface properties.
- Assist in signalling and communication.

4. Water and associated ions

Water and ions are not considered the principal structural constituents of the lipid bilayer, but they are important for membrane-associated processes.

22. Write an essay on different types of point mutations.

A point mutation is a change involving a single nucleotide position in DNA. It may involve the substitution of one nucleotide by another or, in a broader usage, the insertion or deletion of a single nucleotide. Point mutations may occur spontaneously during DNA replication or be induced by mutagens such as radiation and chemicals.

Point mutations can be classified according to the type of nucleotide change and their effect on the encoded protein and orgin.

1. Based on the type of nucleotide change mutation can be

Base substitution mutations: If a base substitution occurs when one nucleotide is replaced by another. They can be either **transition mutation** or **Transversion mutation**.

A transition is the substitution of one purine by another purine or one pyrimidine by another pyrimidine. (A → G, G → A, C → T, T → C). Transitions may alter codons, change amino acids or produce no change in the encoded amino acid. They are among the common types of point mutations.

A transversion is the substitution of a purine by a pyrimidine or a pyrimidine by a purine (A → T, A → C, G → T, G → C). The reverse substitutions are also transversions. Transversions can alter codons and may affect protein structure or function. Their effects depend on the location and nature of the substitution.

2. Based on the effect of mutations on protein synthesis mutations can be

A. Silent mutation

A silent mutation changes a codon without changing the amino acid it encodes. Silent mutations often have little effect on protein sequence, although some can influence splicing, mRNA stability or translation efficiency.

Example: In the coding DNA strand, a change from GAA to GAG changes the codon but both specify glutamic acid in the standard genetic code.

B. Missense mutation

A missense mutation changes a codon so that a different amino acid is incorporated into the protein. Significance: Missense mutations may alter protein folding, stability or function. Their effects range from negligible to severe.

Example: In sickle-cell disease, a mutation in the β-globin gene changes the glutamic acid codon to a valine codon in the β-globin protein.

C. Nonsense mutation

A nonsense mutation converts an amino acid codon into a stop codon (UAA, UAG and UGA). Translation may terminate prematurely, producing a shortened protein that is often non-functional. Some affected mRNAs are also degraded by cellular quality-control mechanisms.

D. Frameshift mutations

A frameshift mutation occurs when a nucleotide insertion or deletion changes the reading frame of a coding sequence. Such a mutation generally causes a frameshift when the number of inserted or deleted nucleotides is not a multiple of three.

Example:

In an mRNA with original coding sequence: AUG-AAA-GGC-UUU if deletion of one nucleotide occurs at the 5' end the coding sequence will be UGA- AAG-GCU- UU

After a one-nucleotide insertion or deletion, the grouping of downstream codons changes.

Frame shift mutation can

- Can alter the downstream amino acid sequence.

- May introduce a premature stop codon.
- Often produces a non-functional protein.

3. Based on origin mutations can be

Spontaneous and induced point mutations

Spontaneous mutations: Arise naturally due to replication errors, spontaneous chemical changes in bases or errors in DNA repair.

Induced mutations: Result from exposure to mutagens, including certain chemicals and ultraviolet or ionizing radiation.

Biological significance of point mutations

Point mutations have several important biological consequences.

- Genetic variation: They introduce new alleles into populations.
- Evolution: Beneficial mutations can contribute to adaptation and natural selection.
- Genetic disorders: Mutations in essential genes may cause inherited diseases.
- Cancer: Mutations in genes regulating cell division, DNA repair and cell survival can contribute to tumour formation.
- Protein function: Mutations may alter protein structure, activity or stability.
- Molecular research: Experimentally induced point mutations help researchers identify the functions of genes and specific amino acids in proteins.

Point mutations are small-scale changes in DNA that may have neutral, harmful or beneficial effects. Transitions and transversions describe the types of base substitutions, while silent, missense and nonsense mutations describe their effects on protein coding. Frameshift mutations caused by single-nucleotide insertions or deletions can also profoundly affect protein synthesis. Point mutations are important in genetic variation, evolution, disease and molecular biology research.